

RUDRA CHAVDA

rudrabchavda@gmail.com | LinkedIn: rudra-chavda | Portfolio: rudra.chavda.online | U.S. Permanent Resident

EDUCATION

University of California, Santa Barbara
Bachelor of Science, Computer Engineering

Expected Spring 2028
GPA: 3.95

- **Honors:** 6x Dean's List
- **Coursework:** Data Structures and Algorithms, C Programming, Computer Architecture, Operating Systems, Software Engineering.

TECHNICAL SKILLS

Languages	Python, C, C++, Java, TypeScript, JavaScript, SQL, Assembly
Frameworks	Vue.js, React, Next.js, Node.js, Py4Web, Tailwind CSS, Angular
Tools	Git, Nautilus (Cluster), CARLA, Docker, GDAL, VS Code, Google Colab, Machine Learning
Concepts	Reinforcement Learning, Data Structures, Algorithms, System Design, Open Source, team building

EXPERIENCE

Ascend Cargo June 2026 – Present
AI/ML Intern

- Built MCP Tools for a custom LLM agent with (**Java 21, Spring Boot, OpenAI function calling**), letting operations staff create and update freight orders up to **300% faster** for customers
- **Designed and shipped an inbound email channel over the Microsoft Graph API** with OAuth client-credentials auth, allowing verified staff to query live shipment and order data by emailing a shared mailbox
- **Hardened agent tool design after live-demo failures: replaced silent fallbacks** (arbitrary tenant selection, defaulting missing timestamps to now()) with explicit clarifying questions, eliminating a class of plausible-looking-but-wrong records.
- **Authored 43 unit tests** covering authorization, tenant isolation, prompt-injection stripping, and serialization, plus operator documentation for the Entra ID and Exchange configuration.

AIEA Lab (UCSC) Dec 2025 – Apr 2026
Undergraduate Researcher

- Wrote and benchmarked Reinforcement Learning (RL) algorithms such as **DQN, A3C**, and **Stable Baselines** in the Car Racing environment to evaluate autonomous vehicle safety as part of the Robustifying AV's project.
- Deploying the **CARLA** simulator in a headless environment on the **Nautilus** hyper-cluster (Kubernetes).
- Analyzing edge-case failures by visualizing agent behaviors and reward functions post-training.

CA-Futures (UCSC EEB & Baskin Engineering) May 2025 – Sep 2025
Software Developer

- Developed a full-stack conservation modeling application using **Vue.js** (frontend) and **Py4Web** (backend).
- Integrated geospatial analysis tools (**GDAL**, Ecoscape) to track species connectivity and visualize ecological data.
- Optimized simulation parameters to process large-scale biological datasets for the Ecology Lab.

Opennote (YC S25) Jul 2025 – Feb 2026
Campus Growth Lead

- Co-organized UCSC Cruz hacks in collaboration with Opennote where we gave out \$1500 worth of prizes.
- Organized workshops talking about user-agent API flow in typescript.
- Spearheaded campus growth strategy, increasing user engagement by 30% and acquiring 50+ active users.

Fassur June 2023 – Oct 2023
Front-End Engineering Intern

- Architected a secure, interactive web application using **React.js** and **Tailwind CSS**.
- Designed a responsive front-end framework to improve data visualization for client dashboards.

PROJECTS

Better Notes | *TypeScript, Tailwind CSS, Vercel*

Spring 2025

- Engineered a full-stack note-taking platform with a custom markdown rendering engine for student study workflows.
- Secured 100+ active users within 3 months of deployment to Vercel.
- Live Demo: betternotes-two.vercel.app

Div Explorer Project | *Machine Learning, Python*

Fall 2025

- Developed actionable data labeling pipelines to detect duplicate discoveries and high-overlap patterns.
- Implemented categorization metrics (interpretability, coverage, anomalies) to improve ML model fairness.

Personal Portfolio | *Next.js, Three.js, GSAP*

Winter 2024

- Built a high-performance portfolio site featuring 3D interactive elements using Three.js and GSAP animations.

LEADERSHIP

Eagle Scout Project

Summer 2024

Project Lead, Troop 537

- Directed a team of 20+ Scouts in the construction of redwood structures for Sycamore Grove Park (Livermore).